09-20 第2篇

阅读理解（说明文） How Does Soap Actually Clean Your Hands?

① Every student is told, **"Wash your hands with soap before you eat!"** Most of us do it without thinking. Yet soap is a kind of **small daily magic** we rarely understand. A bar of white soap, a drop of cheap liquid soap, or the clear bar of a hotel — all of them work on the same simple idea, discovered thousands of years ago and only fully explained by modern chemistry. So, **how does soap actually clean?**

② The main problem is **oil**. Much of the dirt on our hands — food fat, sweat, skin cells, and **microbes** (very small living things such as bacteria and viruses) — is held together by a thin film of oil. Plain water cannot remove this film. You can try it yourself: run cold water over a buttered plate, and the butter simply stays where it is. Water and oil **do not mix**, because water molecules are pulled towards other water molecules, while oil molecules prefer the company of oil.

③ Soap solves this problem by building a **tiny bridge** between water and oil. A single soap molecule has **two different ends**. One end — sometimes called the "**head**" — loves water and hates oil. The other end — the "**tail**" — loves oil and hates water. Scientists call these two ends **hydrophilic** (water-loving) and **hydrophobic** (water-fearing). Imagine a tiny magnet that attracts water on one side and oil on the other. Millions of these tiny two-headed magnets, mixed into a drop of water on your hands, are what actually do the cleaning.

④ When you rub your hands with soap and water, something beautiful happens at a **very small scale**. The hydrophobic tails push themselves into the oily dirt; the hydrophilic heads point outwards into the water. Each little drop of oil becomes a **tiny ball** surrounded by a crown of water-loving heads. Once wrapped like this, the oil drop cannot stick to your skin any longer. When you rinse, the running water simply carries away these little wrapped balls — and with them the dirt, the bacteria, and even many viruses trapped inside.

⑤ This is also why soap works so well against viruses like the **coronavirus**. Many viruses are not "naked" — they wear a thin **outer coat made of oil-like material**. When soap molecules meet that coat, the hydrophobic tails slip into it and **break it apart**, the way a careless finger breaks a soap bubble. Once the coat is broken, the virus falls to pieces and can no longer enter any cell. That is why doctors tell us, firmly: **hand-washing with soap for at least 20 seconds is more powerful than it looks**.

⑥ A few practical lessons follow from this science. First, **warm water** is better than cold, because it helps oils melt a little and mix with the soap faster. Second, **rubbing matters** — it is the mechanical action that breaks up the oil drops and sends them into the water, so you must wash all parts of the hands: palm, back, between fingers, thumbs, and nails. Third, **time matters**: a rushed three-second wash picks up only a small part of the dirt. And fourth, **the kind of soap does not matter as much as people think**. Fancy soap with flower smell, plain white soap, or a simple bar hotel soap all clean equally well, because the tiny two-headed magnets inside them are almost the same.

⑦ Soap is an invention that is **over 4 000 years old**. Ancient peoples in Babylon mixed animal fat with plant ash to make the first cleaning paste. They did not know about hydrophilic heads or hydrophobic tails. But by patient observation, they had found something that — as we now understand — works on the smallest invisible level to keep our bodies and our homes safe. Every time you turn on a tap and rub soap in your hands, you are using one of the **oldest and smartest tools humans ever made**.

注释： microbe 微生物；hydrophilic 亲水的；hydrophobic 疏水的；coronavirus 冠状病毒；Babylon 巴比伦

根据文章内容，选择最佳答案。

1. Why can't plain water wash off the dirt on our hands (Paragraph 2)?

A. Because water is too cold.
B. Because much of the dirt is held by a thin film of oil, and water and oil do not mix.
C. Because water is acidic.
D. Because hands are always dry.

2. What is special about a single soap molecule (Paragraph 3)?

A. It has only one end that loves water.
B. It has two different ends: a water-loving "head" (hydrophilic) and an oil-loving "tail" (hydrophobic), acting like a tiny two-headed magnet.
C. It is round and smooth all over.
D. It becomes active only at night.

3. According to Paragraph 4, what actually happens when you rub soap on your hands and rinse with water?

A. Soap becomes a small stone that scratches the dirt off.
B. Soap makes the hands hotter and burns the dirt.
C. Hydrophobic tails go into oily dirt, hydrophilic heads point outwards; each oil drop is wrapped into a tiny ball that running water carries away.
D. Soap changes the colour of the skin so dirt cannot be seen.

4. Why does soap work well against many viruses such as the coronavirus (Paragraph 5)?

A. Because soap is medicine that kills viruses with heat.
B. Many viruses have a thin oil-like outer coat; soap's hydrophobic tails break that coat apart, and the virus falls to pieces.
C. Because viruses are afraid of soap smell.
D. Because the soap cover forms a hard shield around the hand.

5. Which of the following practical lessons is TRUE according to Paragraph 6?

A. Warm water is better than cold, and rubbing all parts of the hands matters.
B. Three seconds is enough for a good wash.
C. Expensive soap is always much better than cheap soap.
D. People should wash their hands without any water, only with dry soap.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "thin film of oil ... water and oil do not mix"
【解析】油膜使水难以带走污垢。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "two different ends ... hydrophilic ... hydrophobic ... tiny two-headed magnet"
【解析】肥皂分子两端亲疏性相反。A、C、D 均错。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第④段 "hydrophobic tails push ... hydrophilic heads point outwards ... running water carries away"
【解析】油污被包裹成小球被水冲走。A、B、D 均错。故选 C。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段 "thin outer coat made of oil-like material ... hydrophobic tails slip into it and break it apart"
【解析】肥皂分子破坏病毒油状外壳。A、C、D 均错。故选 B。

5.
【答案】A
【题型】细节理解题
【定位】第⑥段 "warm water is better than cold ... rubbing matters"
【解析】温水与全面揉搓都有帮助。B 与 "time matters" 相悖；C 与 "the kind of soap does not matter as much" 相悖；D 无水无法冲走污垢。故选 A。